MYCRO-TEC® DIGESTER

Stubble is slow to break down on its own.

Digester adds four selected decomposer organisms to the residue after harvest. They establish at the damp base of the stubble and start working through the cellulose, hemicellulose, lignin and pectin that hold last season's carbon and nutrients.

USED ON CROP RESIDUE IN



01

Vegetables

Crop trash left after harvest



02

Orchards

Prunings and leaf litter on the orchard floor. Not for nut trees.



03

Broadacre

Cereal, rice, canola and legume stubble

ACROSS THE SEASON

How residue breaks down.

Breakdown depends on moisture and takes time.



01

Establishes at the base

The organisms take hold first where residue meets soil and moisture is held, then spread through damp stubble.



02

Opens the structure

Their enzymes and organic acids work on cellulose, hemicellulose, lignin and pectin, so the carbon locked in the fibre becomes available.



03

Breakdown continues

The selected organisms and the soil's own community keep working through the material ahead of the next crop.

WHAT IS IN IT

Four organisms, declared on the label.

Three fungi and one actinomycete. Counts are colony-forming units per millilitre at production, as declared on the label.

01

Trichoderma harzianum

FUNGUS

$>2\times10^{8}$

CFU/ML

Colonises residue rich in cellulose. Its enzymes begin to break down the fibre.

02

Aspergillus niger

FUNGUS

$>2\times10^{7}$

CFU/ML

Produces organic acids and enzymes that work on the tougher, more lignified material.

03

Streptomyces lydicus

ACTINOMYCETE

$>2\times10^{7}$

CFU/ML

An actinomycete that breaks down residue alongside the fungi.

04

Penicillium radicum

FUNGUS

$>1\times10^{7}$

CFU/ML

Works on the surface of residue and at its damp base, alongside Trichoderma.

WHAT THEY ACT ON

Cellulose

The main plant fibre

Hemicellulose

The supporting fibre network

Lignin

The hard, resistant structure

Pectin

The material that binds plant cells



CHECKING RESIDUE BREAKDOWN

Start at the stubble base.

Check for softer stalks at the damp base and residue that pulls away from the soil more easily. Discuss what you find with your agronomist.

RESIDUE AND APPLICATION CONDITIONS

When to consider Digester.

Talk to your Hybrid-Ag agronomist about the residue left after harvest and the nutrition required for the next crop. They can advise whether Digester and a prescription blend are appropriate.

01

Residue that will still be there at sowing

Cereal, canola, legume, vegetable or orchard residue that would otherwise carry intact into the next crop.

02

Heavy or lignified loads

Big stubble loads, or material that has sat through a dry summer and stalled.

03

Applying close to rain

Applied soon after harvest and as close to rain as you can, before or after it. The breakdown runs on moisture in the residue.

04

Soil and water management

Soil and water conditions affect residue breakdown. Discuss these alongside your nutrition requirements.

Hot, dry spells

Activity falls away through prolonged heat and dryness. Keep out of the heat of the day and line the application up with rain.

Time in the tank

Mix and apply the same day. Overnight in nutrient the fungi grow mycelia and block nozzles.

Nitrogen on its own

Nitrogen feeds the biology already there; it does not add decomposers. The 8 kg of ammonium sulphate is the starter. Do not add more nitrogen to the mix.

REFERENCE

Rates, mixing and handling.

Application rates, mixing, storage and handling.

CROP GROUP	DIGESTER	STARTER	WATER
01 **Vegetables** Crop trash left after harvest	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water
02 **Orchards** Prunings and leaf litter on the orchard floor. Not for nut trees.	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water
03 **Broadacre** Cereal, rice, canola and legume stubble	**1–2 L/ha** up to 3 L/ha heavy residue	**8 kg/ha** every application	**100 L/ha** 1:50 product : water

Label rates. The ammonium sulphate stays at 8 kg/ha whatever the residue load. One application after harvest; not for in-crop use.

01

Mixing

Shake the drum well. Add Digester to cool, clean, non-chlorinated water first (neutral pH, not below 5.5), then add the ammonium sulphate and stir thoroughly.

Strain as you pour. Keep nozzles and filters at 150 µm or larger.

Mix and apply the same day. Left in the tank overnight it grows mycelia and blocks nozzles.

Digester, water and ammonium sulphate only. Do not swap the starter for urea, fish or anything else.

02

Storage

Store at 5–40 °C in a clean, dry place out of direct sunlight, drums upright and closed. Do not freeze.

Twelve months from the date of manufacture. Once a drum is open, use it rather than keep it: the organisms start to germinate and expire without nutrient. Never store the diluted mix.

Growth on top of the liquid is mycelia, a sign the biology is alive. Shake and use as normal.

03

Safety

Not classified as hazardous. Keep out of reach of children; may irritate skin or eyes, so wash after handling.

Wear gloves and eye protection when mixing. Follow the current label and safety data sheet.

Not certified organic. Allowable-input status is assessed case by case with your certifier.

TALK TO US

Talk to us about Digester.

Tell us the crop, the residue load and when you expect rain. Your Hybrid-Ag agronomist can advise on timing and rate. Label and safety data sheet available on request.

(03) 5722 7555

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Mycro-Tec® Digester is made in Australia by Great Southern Biology and supplied by Hybrid-Ag.